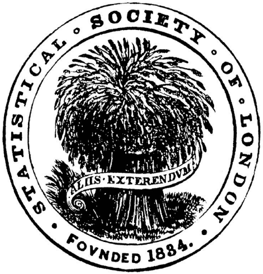
What is statistics?

Presentation Outline

Origins

Originally, collections of data on the condition of the state

- One of the oldest statistical societies:
  - 1834: Royal Statistical Society founded (Great Britain)
- RSS founding aim: "the collection and classification of all facts illustrative of the present condition and prospects of Society, $\cdots$"
  - Members called themselves "statists"
  - Seal was, and still is, a wheat sheaf
  - Original motto was *"Aliis Exterendum"* (for others to thresh out, i.e. interpret)



Origins - 2

- On this side of the Atlantic:
  - 1839: American Statistical Association founded (USA)
- ASA aims were similar. Early ASA members included
  - President Martin Van Buren
  - Florence Nightingale
- Mathematical ideas essentially absent from either society until 1880's
- Became more frequent in both societies in the 1930's
- Both societies now focus on developing and evaluating statistical methods

Dichotomy between collecting and analyzing data

- Still persists today
  - University libraries have thousands of books like "Agricultural Statistics, Iowa (1994)"
  - hundreds on methods to analyze data
- US Department of Agriculture has two job titles for statisticians
  - Agricultural or Survey Statistician: collect data, do some analysis, BS with 15 hours math/stat
  - Mathematical Statistician: analyze data, develop new methods, BS with 24 hours math/stat, MS or PhD preferred

Definitions of Statistics, as known today

Statistics is:

- A systematic method for drawing conclusions from data with variability
- The scientific study of how to:
  - collect data (design)
  - summarize data (description)
  - draw conclusions and make predictions from incomplete data (estimation, inference, prediction)

A model for statistics in scientific investigations

$$\text{Question} \underset{\text{Design}}{\Longrightarrow} \text{Data} \underset{\text{Analysis}}{\Longrightarrow} \text{Conclusion}$$

- Why does the question matter?
- Why does the design matter?

Five components

design, description, estimation, inference, prediction

1. Design: How to collect data (not a focus in this class)
2. Description: describing patterns in the sample
3. Estimation: estimating population quantities from sample statistics
4. Inference (p 8): "conclusion that patterns in the data are present in some broader context"
5. Statistical Inference:
   "above justified by a probability model linking the data to a broader context"
6. Prediction: predicting characteristics, e.g., values of new observations

Important points:

- Statistics is using data to answer questions
- Five components: design, description, estimation, inference, and prediction